

Linux Reinstall Grub 2 with a Linux Live USB  
or other Linux live media

# Reinstall Grub2 on UEFI / EFI systems

## Reinstall Grub2 to the EFI partition using chroot (change root)

### On EXT4 & BTRFS Filesystems

How to Repair, Restore, or Reinstall Grub 2 with a Linux Live  
CD or USB

For various reasons: it happens the bootloader get's corrupted, erased or misconfigured resulting in failure to boot. To restore system operation without re-installing your OS or losing your data you will need to Reinstall (Repair/Restore) Grub2 using a Linux live USB. (or other external Linux live media)

**! NOTE: How Bootable Devices Work in UEFI Boot Mode: (If efi /esp boot partition is not detected)**

**NOTE:** UEFI firmwares are not implemented consistently across manufacturers.

Some BIOS/UEFI firmwares must have (EFI/boot/bootx64.efi) in the root of a fat32 formatted device/partition, in order to detect the efi partition, and thus for BIOS to show a UEFI boot option for the Device.

•we can create this file in the in the directory path <EFI/boot/bootx64.efi> manually, or copying and renaming the created .efi file for linux to boot in the directory <EFI/boot/bootx64.efi>

•or we can add the --removable option to grub-install to create this file if the efi partition is not shared with windows (or another OS, because it will overwrite any previous .efi file located in that directory <EFI/boot/bootx64.efi> possibly rendering the other OS unable to boot)

# Reinstall Grub2 on EXT4 filesystem. EFI

## Terminal Commands

Mount the partition your Linux Installation is on. If you are not sure which it is, launch GParted (included in the Live USB) or `lsblk` or `lsblk -f` or `fdisk -l` commands. Replace the XY with the drive letter, and partition number, for example: `sudo mount /dev/sda2 /mnt.`

```
sudo mount /dev/sdXY /mnt
```

mount the efi partition to the `/boot/efi` directory of the linux installation

```
sudo mount /dev/sdXY /mnt/boot/efi
```

Now bind the directories that grub needs access to to detect other operating systems, like so.

```
sudo mount --bind /dev /mnt/dev
sudo mount --bind /dev/pts /mnt/dev/pts
sudo mount --bind /proc /mnt/proc
sudo mount --bind /sys /mnt/sys
```

copy and paste: `sudo mount -B /dev /mnt/dev && sudo mount -B /dev/pts /mnt/dev/pts && sudo mount -B /proc /mnt/proc && sudo mount -B /sys /mnt/sys`  
\* you can use `-B` instead of `--bind` both are equivalent

Now we jump into that using `chroot`.

```
sudo chroot /mnt
```

Now install, check, and update grub.

Now you install grub to the efi partition, for example: `grub-install /dev/sda1` `grub-install --recheck /dev/sda.`

**\*caution\*** then you may want to use the `--removable` option: `grub-install <efi partition> --removable`

```
grub-install /dev/sdXY
```

```
grub-install --recheck /dev/sdXY
```

```
update-grub
```

Now grub is back, all that is left is to exit the chrooted system and unmount everything.

```
exit
sudo umount /mnt/sys
sudo umount /mnt/proc
sudo umount /mnt/dev/pts
sudo umount /mnt/dev
sudo umount /mnt
```

copy and paste: `sudo umount /mnt/sys; sudo umount /mnt/proc; sudo umount /mnt/dev/pts; sudo umount /mnt/dev; sudo umount /mnt`

Shut down and turn your computer back on, and you will be met with the default Grub2 screen.

**You may need to Remove/Edit the previous EFI partition mount point in /etc/fstab  
If you still encounter problems booting/entering into your Linux System**

**You may need to Edit the /etc/crypttab if the linux system was installed in a partition encrypted with  
LUKS/cryptsetup/dm-crypt:  
Edit the /etc/crypttab with Linux installation's filesystem uuid**

You may want to update grub or re-install within in the Linux system however you like it  
Congratulations, you have just Repaired/Restored/Reinstalled Grub 2 with a Linux Live USB!

# Reinstall Grub2 on BTRFS Filesystem EFI

**BTRFS Filesystem with /@ subvolume**

**:bind mounting subvolumes BTRFS (if @ subvolume  
is present) & reinstall grub2**

On a BTRFS filesystem where the OS is installed in a subvolume (looks like a directory, instead of being in the root of the filesystem), usually this is /@. we need to **bind mount** the subvolume to /mnt,  
or specify the subvol option in the mount command: `mount -o subvol=/@ <path-to-mount-to>`

A **bind mount** takes an existing directory tree and replicates it under a different mount point. The directories and files in the bind mount are the same as the original. Any modification on one side is immediately reflected on the other side, since the two views show the same data.

The **subvol** mount option is used to mount the subvolume of a Btrfs filesystem using its relative path.

## Terminal Commands (for BTRFS with system in a subvolume)

Bind Mount the partition your Linux Installation is on (if within a subvolume). If you are not sure which it is, launch GParted (included in the Live USB) or `lsblk` or `lsblk -f` commands. Replace the XY with the drive letter, and partition number, for example: `sudo mount /dev/sda2 /mnt`.

```
sudo mount /dev/sdXY -o subvol=@ /mnt
```

\* or `sudo mount --bind </dev/sdXY/@`s mount point> /mnt`

mount the efi partition to the `/boot/efi` directory of the linux installation

```
sudo mount /dev/sdXY /mnt/boot/efi
```

Now bind the directories that grub needs access to to detect other operating systems, like so.

```
sudo mount --bind /dev /mnt/dev
sudo mount --bind /dev/pts /mnt/dev/pts
sudo mount --bind /proc /mnt/proc
sudo mount --bind /sys /mnt/sys
```

copy and paste: `sudo mount -B /dev /mnt/dev && sudo mount -B /dev/pts /mnt/dev/pts && sudo mount -B /proc /mnt/proc && sudo mount -B /sys /mnt/sys`

\* you can use `-B` instead of `--bind` both are equivalent

Now we jump into that using `chroot`.

```
sudo chroot /mnt
```

Now install, check, and update grub.

Now you install grub to the efi partition, for example: `grub-install /dev/sda1`  
`grub-install --recheck /dev/sda`.

\*caution\* then you may want to use the `--removable` option: `grub-install <efi partition> --removable`

```
grub-install /dev/sdXY
```

```
grub-install --recheck /dev/sdXY
```

```
update-grub
```

Now grub is back, all that is left is to exit the chrooted system and unmount everything.

```
exit
sudo umount /mnt/sys
sudo umount /mnt/proc
sudo umount /mnt/dev/pts
sudo umount /mnt/dev
sudo umount /mnt
```

copy and paste: `sudo umount /mnt/sys; sudo umount /mnt/proc; sudo umount /mnt/dev/pts; sudo umount /mnt/dev; sudo umount /mnt`

Shut down and turn your computer back on, and you will be met with the default Grub2 screen.

**You may need to Remove/Edit the previous EFI partition mount point in /etc/fstab  
If you still encounter problems booting/entering into your Linux System**

**You may need to Edit the /etc/crypttab if the linux system was installed in a partition encrypted with  
LUKS/cryptsetup/dm-crypt:  
Edit the /etc/crypttab with Linux installation's filesystem uuid**

You may want to update grub or re-install within in the Linux system however you like it

Congratulations, you have just Repaired/Restored/Reinstalled Grub 2 with a Linux Live USB!